

PROJECT HANDOVER DOCUMENT — KOYONZO FAMILY CARE CLINIC SYSTEM

1. PROJECT OVERVIEW

A clinic management system that handles:

- Patient registration
- Visits & triage
- Doctor consultations
- Prescriptions & pharmacy dispensing
- Billing & receipts
- Appointment scheduling with reminders
- Management reports for the CEO/Director

Current State: Built in Microsoft Access (Desktop). Migrating to a web-based system for multi-user, remote access.

2. USER ROLES

Role Responsibilities

Receptionist (Staff) Register patients, create visits, view appointments, print receipts

Triage Nurse Enter vitals (BP, Temp, Pulse, Weight, Height, BMI, Chief Complaint, Lab Tests)

Doctor View waiting list, add consultation notes, prescribe drugs, set next appointment

Pharmacist Dispense drugs, mark as billed

Cashier Generate receipts, accept payment (Cash / M-PESA / Card / Insurance)

Administrator Manage users, drugs, services, prices, visit types

CEO / Director View management reports and dashboards

3. DATABASE STRUCTURE

3.1 — Patients

Field Type Notes

PatientID AutoNumber Primary Key

FullName Text(100)

SerialNumber Text(20) Format: PAT-YYYY-####

Phone Text(20)

Gender Text(10) Male / Female

DOB Date Used to calculate age

Insurance Text(50) NHIF / AAR / Private

NOKContact Text(50) Next of Kin contact

RegisteredDate Date/Time Auto-set on registration

3.2 — Visits

Field Type Notes

VisitID AutoNumber Primary Key
PatientID Number FK → Patients
VisitDate Date/Time
VisitType Text(20) Outpatient / Inpatient / Consultation
VisitService Text(50) e.g., General Consultation
Doctor Text(50)
TriageDone Yes/No
ConsultationDone Yes/No
Completed Yes/No
PrimaryServiceID Number FK → tblServices
VisitService Text(50)

3.3 — Triage

Field Type
TriageID AutoNumber (PK)
VisitID Number (FK)
ChiefComplaint Text(255)
LabTests Text(255)
LabStatus Text(20)
BP Text(20)
Temp Number
Pulse Number
Resp Number
Weight Number
Height Number

BMI Number

3.4 — Consultation

Field Type

ConsultationID AutoNumber (PK)

VisitID Number (FK)

History Text(500)

Examination Text(500)

Diagnosis Text(500)

Treatment Text(500)

Referral Yes/No

Facility Text(100)

3.5 — Prescriptions

Field Type Notes

PrescriptionID AutoNumber PK

VisitID Number FK → Visits

Drug Number FK → tblDrugs

Quantity Number

Dosage Text(50) e.g., 500mg

Frequency Text(50) e.g., Twice a day

Duration Text(50) e.g., 5 days

Status Text(20) Pending / Dispensed

NextAppointment Date Follow-up date

3.6 — tblDrugs (Master Drug List)

Field Type

DrugID AutoNumber (PK)

DrugName Text(100)

UnitPrice Currency

CurrentStock Number

3.7 — tblDispensed (Bridge — Handover)

Field Type

DispensedID AutoNumber (PK)

PrescriptionID Number (FK)

ItemID Number (FK → tblDrugs)

VisitID Number (FK)

PatientID Number (FK)

Quantity Number

DispensedDate Date/Time

DispensedBy Text(50)

IsBilled Yes/No

3.8 — tblReceipt (Billing Header)

Field Type

ReceiptID AutoNumber (PK)

ReceiptNo Text(20)
PatientID Number (FK)
ReceiptDate Date/Time
SubTotal Currency
VATAmount Currency
DiscountAmount Currency
TotalAmount Currency
PaymentMethod Text(20)
TransReference Text(50)
IsPaid Yes/No
ServedBy Text(50)

3.9 — tblBilling (Billing Lines)

Field Type
BillingID AutoNumber (PK)
ReceiptID Number (FK)
ItemID Number (FK → tblDrugs or tblServices)
Quantity Number
UnitPrice Currency
LineTotal Currency
TransDate Date/Time

3.10 — tblServices (Master Services List)

Field Type

ServiceID AutoNumber (PK)

ServiceName Text(100)

ServiceCategory Text(50)

ServiceCost Currency

IsActive Yes/No

4. THE COMPLETE USER WORKFLOW

STEP 1: Patient Arrives

- Receptionist opens the system → New Patient Registration
- Enters: FullName, Phone, Gender, DOB, Insurance, NOKContact
- System generates Serial Number automatically (Format: PAT-YYYY-####)
- System creates a Visit for today

STEP 2: Triage

- Nurse opens the Triage screen
- Enters: BP, Temp, Pulse, Resp, Weight, Height, Chief Complaint, Lab Tests
- System calculates BMI
- Marks TriageDone = True

STEP 3: Doctor Consultation

- Doctor opens the Doctor's Dashboard
- Sees the Waiting List (patients with TriageDone = True and ConsultationDone = False)
- Selects a patient
- The VisitID field blinks (yellow) to confirm it is linked to the correct patient
- Enters: History, Examination, Diagnosis, Treatment
- Prescribes drugs (multiple can be added)
- Sets Next Appointment (optional)
- Clicks Complete Visit → marks ConsultationDone = True

STEP 4: Pharmacy

- Pharmacist opens the Prescription Queue
- Sees all prescribed drugs for today
- Confirms which drugs were given → writes to tblDispensed

STEP 5: Cashier

- Cashier opens the Billing Form
- Selects the patient
- Clicks Import Items from Dispensary
- A dialog shows all unbilled items with numbers:

...

1 | Thomas Wesonga | Ceftriaxone 1g | Qty: 5

2 | Thomas Wesonga | Panadol 500mg | Qty: 24

3 | Thomas Wesonga | General Consultation | Qty: 1

...

- Types 1, 2, 3 → OK
- System inserts into tblBilling, looks up prices from tblDrugs, calculates LineTotals
- Marks IsBilled = True on source rows
- Cashier selects Payment Method (Cash / M-PESA / Card / Insurance)
- Prints receipt

STEP 6: Management Reports

- CEO opens the Management Report dialog
- Types a number (1–10) for the report they want:

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1 = Drug Summary

2 = Visitor List

3 = Admitted Patients

4 = Patients by Service Category

5 = Top 10 Most Prescribed Drugs

6 = Doctor Activity

7 = Visit Types Distribution

8 = Lab Tests Summary

9 = Revenue Snapshot

10 = All Reports Combined

...

- Report is generated and copied to clipboard / saved as Excel

STEP 7: Appointments

- Receptionist or Doctor clicks Appointments

- Types a number (1–6):

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1 = Tomorrow

2 = Next 5 days

3 = Upcoming (30 days)

4 = Missed (60 days)

5 = All

6 = Custom range

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- System shows: Upcoming / Attended / Missed

- Reminder: When the doctor opens the dashboard, a popup shows patients with appointments 2 days from now

5. KEY BUSINESS LOGIC

5.1 — Serial Number Format

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PAT-YYYY-####

Example: PAT-2026-0001

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Generated automatically on the first save.

5.2 — Age Calculation

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Age = DateDiff(yyyy, DOB, Today) - 1 if birthday has not passed this year

...

5.3 — Appointment Attendance Detection

- If appointment date is in the future → Upcoming
- If appointment date is in the past AND a Visit exists within 3 days → Attended
- If appointment date is in the past AND no Visit exists → Missed

5.4 — Drug Pricing

- Each drug has a price in tblDrugs.UnitPrice
- When dispensed, price is looked up and stored in tblBilling.UnitPrice
- LineTotal = Quantity × UnitPrice

5.5 — VAT

- Applied at 16% (configurable)
- SubTotal + VAT - Discount = TotalAmount

5.6 — Ghost Record Prevention

- A consultation record is only saved if History OR Examination is not empty
- Otherwise, the save is silently cancelled

6. THE KEY FEATURES (For the Developer)

Feature Description

Waiting List Popup showing patients who have been triaged but not yet consulted

Blinking VisitID Visual indicator in the consultation form that the form is linked to the correct patient

Set Next Appointment Doctor sets follow-up date from the prescription form

Appointment Reminders Auto-popup on Doctor's Dashboard showing appointments in 2 days

Drug Import Bridge Button on Billing Form that pulls dispensed drugs into the receipt

Number-Based Dialogs Simple input (type 1, 2, 5) for reports and imports — no complex dropdowns

Clipboard Output Reports can be copied as plain text for WhatsApp/Email

Excel Export Reports can be exported to Excel

PDF Receipts Receipts can be printed as PDF

Thermal Printer Support 80mm thermal receipt printer (X Printer 80C)

Management Reports 10 different reports for the CEO

No Cost Prescription Slip Medical document printed for the patient (no prices shown)

Cost Receipt Financial document printed for the patient (with prices, VAT, total)

7. NON-FUNCTIONAL REQUIREMENTS

Requirement Value

Multi-user Yes — multiple staff using at once

Remote Access Optional — via web browser

Authentication Username + Password per user

Role-Based Access Yes — different menus per role

Audit Log Log who did what and when

Backup Daily automatic backup

Data Retention 7 years (medical records)

Response Time Page load < 2 seconds

Mobile-Friendly Optional — for doctors on tablets

8. So this is the summary of what you are to build

8.1 — Pages / Screens

1. Login — Username + Password
2. Dashboard — Stats: patients today, waiting, cleared, revenue
3. Patient Registration — New patient form
4. Patient Search — By name, phone, serial
5. Visit Management — Create, view visits
6. Triage Entry — Vitals + chief complaint
7. Doctor's Dashboard — Waiting list + consultation + prescriptions

- 8. Pharmacy — Dispense queue
- 9. Billing / Receipt — Header + line items
- 10. Appointments — Set, view, remind
- 11. Reports — 10 management reports
- 12. Admin — Users, drugs, services, prices

- 1. Review this document. Confirm every field and workflow is correct.
- 2. Gather screenshots from your Access system.
- 3. Clarify your budget and timeline.
- 4. Send the package to the developer.
- 5. Schedule a kickoff meeting to walk through the workflows.
- 6. Sign a clear contract covering ownership, support, and data protection.

This document is your blueprint. Save it, print it, and share it with confidence.

If you want, I can also produce:

- A one-page summary for the developer (elevator pitch).
- A detailed requirements spec (per screen with exact fields).
- A sample contract template.

Just tell me which one you want next.